

SHANMUGARAJ

Full-Stack Developer (MERN) | Software Application Development

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SUMMARY

Full-stack developer (MERN) building and deploying end-to-end applications with React.js, Node.js, PostgreSQL, and MongoDB. Experienced in designing RESTful APIs, JWT-based authentication, and role-based access control (RBAC), with hands-on use of AI-assisted development workflows (Claude, Claude Code) for faster debugging and code review. Strong CS foundation — data structures, databases, operating systems, computer networks — with 150+ LeetCode problems solved. 2025 ECE graduate who self-transitioned into software development

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, C++, SQL, Python

Backend & AI: Node.js, Express.js, REST APIs, Prisma ORM, Redis, RAG basics, Prompt Engineering

Databases: PostgreSQL, MongoDB

Developer Tools: Git, GitHub, Postman, Vercel, Docker/CI-CD

EDUCATION

B.E., Electronics & Communication Engineering

2021 - 2025 (Graduated)

Jerusalem College of Engineering, Chennai

CGPA: 7.5/10 | No Arrears | Coursework: DSA, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Software Intern | NSIC - Technical Services Center

2024

- Developed and debugged C/C++ applications for microcontroller platforms, building strong foundations in low-level programming, memory management, and systematic root-cause analysis.
- Integrated AI/ML inference into embedded systems and performed structured testing and validation, applying the same debugging discipline to full-stack project development.

PROJECTS

Tiny-URL Scalable URL Shortener | *Authorization Node.js · Express.js · MongoDB · Redis · REST API · MVC · Caching*

[GitHub](#)

- Engineered a high-throughput URL shortening microservice using Node.js, Express, and MongoDB, implementing Base62 encoding to generate unique, collision-resistant short codes
- Architected a Redis caching layer using the Cache-Aside pattern, serving 90%+ of read traffic directly from memory and reducing database lookup latency from ~120ms to <10ms.
- Implemented JWT-based URL authorization and route-level middleware to secure endpoint creation, restrict link management to verified owners, and prevent unauthorized access.

Task Flow - Team Task Manager | *React.js, Node.js, Express.js, PostgreSQL, Prisma ORM, JWT, RBAC*

[Live Demo](#) | [GitHub](#)

- Built and deployed a full-stack task management platform with a Kanban board, task filtering, overdue task tracking, and role-based collaboration for team project management.
- Designed MVC RESTful APIs with JWT authentication and admin/member RBAC; optimized PostgreSQL queries using Prisma ORM to improve performance and scalability.
- Kept the codebase clean and production-scalable using modular React components and Git feature branching.

College Transport Tracking System | *Flutter, Dart, Google Maps API, REST API, Real-Time Geolocation*

- Independently learned Flutter/Dart to build a cross-platform live vehicle-tracking app using the Google Maps API; diagnosed and fixed real-time map lag by optimizing REST API polling frequency.

PROBLEM SOLVING

- Solved 150+ LeetCode problems across Arrays, Strings, Trees, Graphs, and Dynamic Programming, with strong command of time and space complexity analysis applied to backend optimization.
- Studying distributed systems and system design fundamentals: scalability, caching, load balancing, database sharding, microservices.

ACHIEVEMENTS

- 2nd Place, College Web Development Competition - built a fully responsive, functional web app recognized among all competing teams.
- Deployed 2 live full-stack applications on Vercel featuring RBAC, JWT authentication, and both SQL and NoSQL databases.